

Coupling Constants as Projection-Stability Parameters: Why α Is Small and α_s Is Large

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Abstract

The fine-structure constant $\alpha \approx 1/137$ and the QCD coupling $\alpha_{s(M_Z)} \approx 0.118$ are conventionally treated as free parameters. We show that within the K - Q projection framework, both are constrained by **projection stability** — the requirement that $Q \circ K$ produce observables whose locality (exponential Green’s-function decay) survives radiative corrections. The derivation chain from gauge group G to coupling bounds uses no hand-tuned parameters: G fixes the Casimir invariants, which fix β_0 , which via dimensional transmutation determines the confinement scale Λ , which enters the locality bound. For QCD, substituting transmutation into the locality bound yields a self-consistency equation that is **automatically saturated** — the coupling is forced to a fixed point, not chosen from an interval. For QED, the abelian structure ($C_A = 0$, $\pi_3(U(1)) = 0$) eliminates both asymptotic freedom and the instanton bound, leaving only the locality constraint; the remaining freedom is the mass hierarchy (an open problem, not a deficiency). The hierarchy $\alpha_{\text{QED}} \ll \alpha_s$ is a topological consequence of $\pi_3(U(1)) = 0$ versus $\pi_3(\text{SU}(3)) = \mathbb{Z}$. We further establish that the Picard–Lefschetz thimble decomposition of K is an instance of the spectral comb ($H = dI + A$, A skew-symmetric), with unitarity providing the antisymmetry constraint. Via this isomorphism, the self-consistency fixed point is shown to be the unique stable solution within its kernel class (Banach contraction), and thimble dominance is upgraded from a numerical bound to a structurally protected spectral gap. All 57 results are Z3-verified.

Keywords: fine-structure constant, coupling constant, QCD, QED, projection stability, beta function, dimensional transmutation, Casimir operator, instanton, ADHM, self-consistency, mass gap, projected ontology, K-Q framework, formal verification, Z3

1 Introduction

The Standard Model of particle physics has 19 free parameters [1]. Among them, the gauge couplings — the fine-structure constant $\alpha \approx 1/137$, the QCD coupling $\alpha_{s(M_Z)} \approx 0.118$, and the weak mixing angle θ_W — are measured to extraordinary precision but not derived from the theory. The question ‘Why does α have the value it does?’ has motivated decades of speculation, from Eddington’s combinatorial attempts [2] through Wyler’s geometric formula [3] to modern anthropic reasoning [4].

This paper offers a different answer within the K - Q projection framework developed in Volumes I–XII of this series [5–16]. The central claim is:

Coupling constants are projection-stability parameters. Their values are constrained by the requirement that the composed map $Q \circ K$ produce observables robust under radiative corrections.

The claim has three components:

1. **Derivation:** Every parameter in the constraint chain is derived from gauge group data (G , R , N_f) via representation theory. No parameter is hand-tuned.
2. **Fixed point:** For QCD, the locality bound and dimensional transmutation form a self-consistency loop whose solution is the physical coupling. The coupling is determined, not chosen.
3. **Structural prediction:** The ratio $\alpha_{\text{QED}}/\alpha_s \approx 1/16$ is a topological consequence of $C_{A(U(1))} = 0$ versus $C_{A(\text{SU}(3))} = 3$, not a coincidence.

Underlying these three components is a deeper principle that recurs throughout the K-Q program:

Locality is not a primitive assumption but a stability condition on the projected kernel. Observable geometry — the exponential decay of Green’s functions, the spectral gap of the propagator, the finiteness of correlations at spacelike separation — is reconstructed as stable coherent behavior of $Q \circ K$, not postulated as fundamental ontology.

In standard QFT, locality is an **input**: one writes local Lagrangians and assumes microcausality. In the K-Q framework, locality is an **output**: it is the condition under which the composed map $Q \circ K$ produces observables with finite spatial support. The coupling constant enters because radiative corrections threaten this condition — too-large corrections destroy the spectral gap that defines exponential decay. The locality bound is therefore not merely a ‘perturbativity condition’ but the requirement that **observable geometry survive quantization**.

The paper does **not** claim to derive $\alpha = 1/137$ from first principles. That would require solving the hierarchy problem — why $m_e/\Lambda_{\text{GUT}} \approx 10^{-14}$ — which remains open. What the paper **does** claim is that the framework separates what is determined (QCD coupling, coupling ratios) from what remains free (mass hierarchy), and provides the structural mechanism (projection stability) that constrains both.

All 24 structural results are machine-verified by the Z3 SMT solver via the Kleis formal verification language [17]. The theory file `theories/pot_alpha_stability.kleis` is the executable companion to this paper.

2 Prior Attempts: A Taxonomy

The question ‘Why 1/137?’ has a long and instructive history. We organize prior approaches into seven categories, ordered by increasing proximity to the present work.

2.1 Numerological approaches

Arthur Eddington (1929–1944) claimed that $\alpha^{-1} = 136$ from counting electron-pair states, later revised to 137 when experiment disagreed [2, 18]. Wolfgang Pauli called this ‘complete nonsense’ and ‘romantic poetry’ [19]. Armand Wyler (1969) derived $\alpha = (9/8\pi^4)(\pi^5/2^4 \cdot 5!)^{1/4} \approx 1/137.036$

from volumes of bounded symmetric domains associated with conformal groups [3]. Adler termed this ‘a number in search of a theory’ [20]. Both approaches produce formulas without identifying a physical mechanism that **requires** the value. We make no such claim: the K-Q framework derives **bounds**, not exact values.

2.2 Anthropic constraints

Barrow and Tipler [4] argued that α must permit stellar nucleosynthesis (specifically, carbon production via the triple-alpha process). Modern lattice effective field theory [21] shows this constraint is loose: α could vary by approximately 4% and still allow carbon-based life. The anthropic bound is biological selection, not physics. Our locality bound is a structural property of the projection Q , independent of observers.

Moreover, the anthropic picture implies a **fragile** universe — constants balanced on a knife-edge that happens to permit life. The K-Q picture implies the opposite: a **robust** universe. The self-consistency fixed point is an attractor (DED-5, DED-7): perturbations of the coupling decay back. The constants are not fine-tuned **for** anything; they are **at** a structurally stable configuration **from which** observable geometry (locality, spectral gaps, finite correlations) follows. Robustness is not anthropic selection; it is projection stability.

2.3 Asymptotic safety

Weinberg [22] proposed that quantum gravity might be nonperturbatively renormalizable if couplings reach a non-trivial UV fixed point. Reuter [23] established evidence for such a ‘Reuter fixed point’ in the Einstein-Hilbert truncation. This program addresses UV completion at the Planck scale. Our work addresses IR self-consistency at the confinement scale. The two programs operate in complementary regimes and do not conflict.

2.4 Naturalness

‘t Hooft’s naturalness criterion [24] states that a parameter is natural if setting it to zero increases the theory’s symmetry. This addresses **mass** hierarchies: the electron mass is natural (chiral symmetry restored as $m_e \rightarrow 0$), but the Higgs mass is not. Naturalness does not address gauge coupling **values** — it explains why $m_e/\Lambda_{\text{Planck}}$ might be small, not why α is small. Our work directly addresses coupling values through a different mechanism: projection stability.

2.5 QED triviality and the Landau pole

Landau (1955) [25] showed that the QED coupling diverges at a finite scale $\Lambda_{\text{LP}} \sim m_e \cdot \exp(3\pi/2\alpha) \approx m_e \cdot \exp(645)$. Lattice simulations confirm that QED is ‘trivial’ — the continuum limit has zero coupling [26]. Our locality bound $\alpha < 2\pi/(|\beta_0| \cdot \ln(\Lambda^2/m^2))$ is **tighter** than the Landau pole bound and has direct physical content: it ensures that finite radiative corrections (which in the K-Q framework are convergent, not divergent [9]) do not destroy the spectral gap of the propagator.

2.6 Banks-Zaks fixed point

Banks and Zaks [27] showed that for N_f just below the asymptotic freedom bound ($N_f < 11N_c/2$), the two-loop beta function has a zero at a perturbative coupling $\alpha_* = -\beta_1/\beta_2$. At this IR fixed point, the coupling is **determined by the theory**. Real QCD ($N_f = 6$ for SU(3)) is far below the conformal window ($N_f \approx 12\text{--}16$), so the Banks-Zaks mechanism does not directly apply. However, it demonstrates a crucial point: gauge couplings **can** be fixed-point values determined by self-consistency, not free parameters.

2.7 Multiple Point Principle

Bennett and Nielsen [28] proposed that Nature adjusts coupling constants to values where multiple vacuum states are degenerate — analogous to the triple point of water. This ‘Multiple Point Principle’ (MPP) predicted the Higgs mass $m_H \approx 135$ GeV [29] before the LHC discovery ($m_H = 125.1$ GeV), and reproduces Standard Model gauge couplings at the Planck scale to $\sim 7\%$ accuracy [30]. MPP is the **closest predecessor** to the present work. Both claim that couplings sit at phase-transition boundaries. The difference: MPP **postulates** this as a principle; we **derive** it as a consequence of projection stability in the K-Q framework. The mechanism is: the coupling must sit at the boundary of the admissible interval because projections that are too weak (small α) produce trivial observables, while projections that are too strong (large α) destroy the spectral gap that defines locality.

3 The Derivation Chain

We now present the complete chain from gauge group data to coupling constraints. Every step is either a theorem of representation theory or a standard result of perturbative QFT, reinterpreted within the K-Q framework. No parameter is chosen or fitted.

3.1 Step 1: Casimir invariants from the gauge group

For a simple gauge group G with generators T_R^a in representation R :

$$C_2(R)\delta_{ij} = T_R^a T_R^a, \quad T(R)\delta^{ab} = \text{Tr}_{R(T^a T^b)}.$$

For SU(N): the adjoint Casimir is $C_A = N$, the fundamental trace index is $T_f = 1/2$, and the fundamental Casimir is $C_F = (N^2 - 1)/2N$. For U(1): $C_A = 0$ (abelian — no self-interaction). These are **fixed by the Lie algebra**, with zero freedom (Z3-verified: ALPHA-1 through ALPHA-3).

3.2 Step 2: The one-loop beta coefficient

The universal one-loop formula [31, 32]:

$$\beta_0 = (11/3)C_A - (4/3)T_f N_f$$

where N_f is the number of active fermion flavors. This coefficient determines whether the theory is asymptotically free ($\beta_0 > 0$) or infrared-free ($\beta_0 < 0$). For QCD (SU(3), $N_f = 6$): $\beta_0 = 7$. For QED

($U(1)$): $\beta_{0, \text{QED}} = -4/3$ per charged lepton. The sign difference — which determines the **entire qualitative behavior** of the running coupling — follows from C_A alone (Z3-verified: ALPHA-4 through ALPHA-6).

In the K-Q framework, β_0 has a structural interpretation. Volume VI [10] showed that ghosts (Faddeev-Popov fields in $\ker(Q)$) contribute to the beta function through the composed map $Q \circ K$. The $11C_A/3$ term in β_0 is the ghost-mediated null-space contribution. Asymptotic freedom is a **consequence of null-space activity**: the gauge structure’s non-abelian character enriches $\ker(Q)$ in a way that screens the coupling at short distances.

3.3 Step 3: The locality bound

The K-Q framework requires that the projection Q preserve locality — observables must have finite support in position space, encoded in the exponential decay of the Green’s function kernel. Radiative corrections at one loop modify the propagator by:

$$\Pi(q^2) \sim (\alpha \cdot |\beta_0| / 2\pi) \cdot \ln(q^2/\mu^2).$$

For the spectral gap to survive (the propagator to decay exponentially at large distances), the vacuum polarization must remain a perturbation:

$$\alpha \cdot |\beta_0| \cdot \ln(\Lambda^2/m^2)/2\pi < 1.$$

This is the **locality bound**: the coupling, the beta coefficient, and the energy hierarchy together determine whether the projection remains stable. For QED with $|\beta_0| = 4/3$ and $\ln(\Lambda_{\text{GUT}}^2/m_e^2) \approx 86$: the bound gives $\alpha < 0.055$. The measured $\alpha = 0.0073$ satisfies this with a factor of 7.5 to spare (Z3-verified: ALPHA-7 through ALPHA-9).

3.4 Step 4: The instanton self-referential bound

For non-abelian gauge groups with $\pi_3(G) = \mathbb{Z}$ (all $\text{SU}(N)$), topologically non-trivial field configurations (instantons) exist. The single-instanton action is $S_1 = 8\pi^2/g^2 = 2\pi/\alpha_s$. The ADHM construction [33] gives the moduli space dimension as $4kN$ for charge- k instantons in $\text{SU}(N)$.

Thimble dominance (Volume XII [16]; upgraded to a structural theorem via the spectral comb isomorphism in Section 5.4) requires that the perturbative saddle dominate the path integral — the multi-instanton sectors must be suppressed. This gives:

$$\alpha^2 \cdot \ln(N_{\text{eff}}) < 2\pi$$

where N_{eff} is the effective number of contributing thimbles. Since $S_1 = 2\pi/\alpha$, the bound is **self-referential**: α determines both the suppression factor and the instanton existence.

For $U(1)$: $\pi_3(U(1)) = 0$, so no instantons exist. The instanton bound is **vacuous** — $U(1)$ has no topological self-consistency constraint. This is a **topological asymmetry** between QED and QCD, arising purely from the homotopy of the gauge group (Z3-verified: ALPHA-10 through ALPHA-12).

3.5 Step 5: Dimensional transmutation and the self-consistency loop

For asymptotically free theories, the running coupling defines a confinement scale:

$$\Lambda = \mu \cdot \exp\left(-2\pi/(\beta_0 \cdot \alpha_{s(\mu)})\right).$$

This is **dimensional transmutation** [34]: the dimensionless coupling is traded for a dimensionful scale. The log ratio appearing in the locality bound is then:

$$\ln(\mu^2/\Lambda^2) = 4\pi/(\beta_0 \cdot \alpha_s).$$

The central observation. Substituting this into the locality bound:

$$\alpha_s \cdot \beta_0 \cdot (4\pi/(\beta_0 \cdot \alpha_s))/2\pi = 4\pi/2\pi = 2.$$

The left-hand side is **identically 2**, independent of α_s , β_0 , or any other parameter. The locality bound is **automatically saturated** by dimensional transmutation. This is not a coincidence — it is a structural tautology of the self-consistency loop:

1. The coupling determines the confinement scale.
2. The confinement scale determines the log ratio.
3. The log ratio enters the locality bound.
4. The locality bound constrains the coupling.
5. The loop closes with the bound saturated.

Interpretation: QCD lives at the **fixed point** of its own self-consistency loop. The coupling is not ‘chosen’ from an admissible interval — it is **forced** to the unique value at which the projection remains self-consistent under dimensional transmutation. This is why $\alpha_{s(M_Z)} = 0.118$ sits at the perturbative boundary: it is the boundary (Z3-verified: ALPHA-16 through ALPHA-18).

4 QED versus QCD: The Topological Asymmetry

The derivation chain produces strikingly different results for the two gauge theories of the Standard Model:

Property	QED ($U(1)$)	QCD ($SU(3)$)
C_A	0 (abelian)	3
β_0	$-4/3$ (not AF)	7 (AF)
$\pi_3(G)$	0 (no instantons)	$\mathbb{Z}$ (infinite tower)
Instanton bound	Vacuous	$\alpha^2 \ln N_{\text{eff}} < 2\pi$
Dimensional transmutation	Landau pole (unphysical)	Confinement scale Λ_{QCD}
Locality bound	$\alpha < 0.055$	Automatically saturated
Measured coupling	$\alpha = 0.0073$ (deep inside)	$\alpha_s = 0.118$ (at boundary)
Status of coupling	Bounded above	Determined
Remaining freedom	Mass hierarchy m_e/Λ	Scheme choice only

The structural prediction is immediate: **QED’s coupling is small because the locality bound is restrictive and the coupling has no mechanism to saturate it. QCD’s coupling is large**

because it is forced to the self-consistency boundary by dimensional transmutation. The ratio $\alpha/\alpha_s \approx 1/16$ is not a numerical coincidence but a topological consequence of $C_{A(U(1))} = 0$ versus $C_{A(SU(3))} = 3$ (Z3-verified: ALPHA-13 through ALPHA-15).

5 Coherence with the POT Program

The α -stability result is not isolated — it connects to and strengthens five earlier volumes of this series.

5.1 Volume VI: The beta coefficient as null-space activity

Volume VI [10] computed $\beta_0 = 11C_A/3 - 4T_F N_f/3$ from convergent Feynman parameter integrals and identified the ghost contribution as null-space activity ($\ker(Q)$ shaping $\text{im}(Q)$ through $Q \circ K$). The present paper takes this one step further: the beta coefficient that Volume VI derives is the **key group-theory input** that determines the locality bound. The derivation chain is: Volume VI computes $\beta_0 \rightarrow \alpha$ -stability shows what β_0 constrains.

5.2 Volume VIII: Promoting Assumption A

Volume VIII [12] presents a conditional reduction of the Yang–Mills mass gap problem. Its single physics input is Assumption A: the gluon anomalous dimension $\gamma > 0$. At one loop, $\gamma_{\text{pert}} = 13N_c/(12\pi) \cdot \alpha_s$. If α_s is **determined** by self-consistency (as the fixed-point result shows), then $\gamma > 0$ is no longer an empirical input — it is a **structural necessity**. The conditional mass gap reduction becomes:

$$\underbrace{A}_{\text{now structural}} + \underbrace{B + C + D}_{\text{math scaffold}} + \underbrace{E}_{\text{QFT bridge}} \Rightarrow \Delta > 0.$$

The remaining gap is only Assumption E (existence of rigorous 4D Yang–Mills theory). The contribution of the present paper to the mass gap program is the promotion of Assumption A from empirical to structural.

5.3 Volume XII: Quantifying thimble dominance

Volume XII [16] showed that the Feynman path integral decomposes into Lefschetz thimbles and that ‘small α implies one-thimble dominance.’ The present paper quantifies this: at $\alpha_s = 0.118$, the single-instanton action is $S_1 = 2\pi/0.118 \approx 53$, giving $Z_1/Z_0 \sim \exp(-53) \approx 10^{-23}$. The perturbative saddle dominates by 23 orders of magnitude. This is not merely a quantitative bound but, as the spectral comb isomorphism shows (next subsection), a **structurally protected** fact: the dominance hierarchy is an attractor (perturbations decay back), the spectral gap $S_1 = 2\pi/\alpha_s$ isolates the perturbative sector, and including higher instantons cannot retroactively destabilize it. The numerical statement 10^{-23} is the **symptom**; the spectral gap is the **cause** (Z3-verified: ALPHA-20, DED-7 through DED-9).

5.4 The spectral comb as canonical form of K

The spectral comb [35] is the operator $H = d \cdot I + A$ where A is real skew-symmetric and tridiagonal. The structure theorem (proved by induction in Lean 4) states: **all eigenvalues of H have $\text{Re} = d$** . This is a theorem of linear algebra, domain-agnostic — it applies to any set of spectral data encoded in the off-diagonals, whether zeta zeros, instanton actions, or any other discrete decomposition.

We now show that the Picard–Lefschetz thimble decomposition of the production kernel K is an **instance** of this universal template. The identification is:

1. **Teeth** of the comb = individual thimble contributions K_σ .
2. **Off-diagonals** = action gaps between adjacent saddles, $\Delta S_{ij} = S_i - S_j = -\Delta S_{ji}$ (antisymmetric by definition).
3. **Integer coefficients** = intersection numbers n_σ (topological).
4. **Antisymmetry constraint** = unitarity (probability conservation). For a unitary evolution $U = \exp(-iHt)$, the generator H is Hermitian; the anti-Hermitian part (the group generator) restricted to real matrices is skew-symmetric. This is the antisymmetry axiom of the spectral comb, **derived from physics** rather than postulated.
5. **Completeness** = the Picard–Lefschetz theorem (the integration cycle decomposes into thimbles with no remainder).

With the isomorphism established, properties **already proved** for the spectral comb transfer to the K–Q kernel. Six deductions are Z3-verified in the companion theory file:

1. **Unique vacuum** (DED-1): The spectral comb map is a contraction ($\|J_F - I\|_F < 1$). By Banach’s theorem, the fixed point is unique. Via isomorphism: the thimble decomposition has a unique stable self-consistent projection vacuum within the given kernel class. (This does not preclude theta vacua or metastable sectors in other kernel classes; it constrains the solution within a fixed topological sector and kernel representation.)
2. **Universal coupling** (DED-2): A non-constant diagonal (different d values for different eigenvalues) is inconsistent with the functional equation. Via isomorphism: different instanton sectors cannot ‘see’ different coupling values. The coupling α is universal across all topological sectors.
3. **Instanton gas converges** (DED-3): The N -truncated spectral comb converges to the full spectrum. Via isomorphism: including finitely many instanton sectors ($k = 1, \dots, N$) converges to the full non-perturbative theory. The dilute instanton gas approximation is structurally justified.
4. **Individual stability** (DED-4): The variance reduction theorem ($\sum \sigma_k = N/2$, $\sum \sigma_k^2 = N/4$ implies $\sigma_k = 1/2$ for all k) transfers: each thimble **individually** sits at the stability boundary, not just ‘on average.’
5. **Dynamical stability** (DED-5): Contraction ($\|J_F - I\| < 1$) means perturbations decay exponentially back to the fixed point. Via isomorphism: the non-perturbative vacuum is dynamically stable — small deformations of the thimble configuration decay rather than grow.
6. **CPT fixes the shift** (DED-6): The functional equation $\xi(s) = \xi(1-s)$ forces $d = 1/2$ for the spectral comb. Via isomorphism: CPT symmetry (CPT maps K to K while exchanging particle and antiparticle sectors) is the physical constraint that fixes the ‘shift parameter’ d . This is the analog of the functional equation in the K–Q domain.

The key consequence for the present paper: deductions 1 and 5 together establish that the self-consistency fixed point (ALPHA-18) is not merely **a** solution of the projection-stability loop, but **the unique stable solution**. Any perturbation of α_s away from the fixed-point value decays back. The coupling is an attractor, not just a boundary value.

Thimble dominance as spectral gap. The isomorphism also upgrades the thimble dominance result of Section 5.3 from a numerical fact to a structural theorem. Three additional deductions (Z3-verified):

7. **Dominance is structurally rigid** (DED-7): Contraction plus vacuum uniqueness imply the dominance hierarchy is an attractor. One cannot continuously deform the system from ‘one thimble dominates’ to ‘all thimbles contribute equally’ — any such perturbation decays back. The ratio $Z_0/Z_1 \sim \exp(S_1)$ is not just large; it is **protected**.
8. **Dominance is a spectral gap** (DED-8): For $SU(N)$, the instanton comb is arithmetic (teeth at $k \cdot S_1$). The spectral bandwidth of the comb operator is proportional to $S_1 = 2\pi/\alpha_s$. Small coupling $\rightarrow$ large $S_1 \rightarrow$ wide spectral gap $\rightarrow$ the perturbative sector is spectrally isolated from the non-perturbative sectors. Thimble dominance **is** the spectral gap of the instanton comb.
9. **Truncation stability protects dominance** (DED-9): Convergence of finite truncation means including higher instantons ($k = 2, 3, \dots$) cannot retroactively change the perturbative sector’s contribution. The perturbative calculation is ‘locked in’ regardless of how many non-perturbative sectors exist. Dominance is protected against contamination from the UV.

The synthesis: in the old picture, thimble dominance was a **numerical fact** ($\exp(-53) \approx 10^{-23}$). In the spectral comb picture, it is a **structural theorem**: the perturbative sector is spectrally isolated, the isolation is dynamically stable (contraction), and including higher sectors cannot destabilize it (truncation convergence). All 33 results are Z3-verified in `theories/thimble_spectral_comb.kleis`.

5.5 Volume IV: Algebraic and dynamical confinement

Volume IV [8] derived confinement as an **algebraic** necessary condition: the non-admissibility of $K_{YM} = dA + A \wedge A$ from the Lie bracket makes color charge fiber-non-invariant. The present paper addresses the **dynamical** complement: why the coupling is **large enough** for the confining potential to trap quarks. The algebraic structure (Lie bracket $\neq 0$) makes confinement **possible**; the self-consistency fixed point makes the coupling **large enough** for it to **occur**. The two levels are complementary, not competing.

6 What Remains Genuinely Free

The derivation chain traces back to three inputs:

1. **The gauge group** G — determines C_A , T_f , $\dim(\text{adj})$, $\pi_3(G)$.
2. **The matter representation** R — determines $T(R)$, $C_2(R)$.
3. **The number of flavors** N_f — from the Standard Model particle content.

From these, everything else is derived. But two quantities are **not** derivable from gauge group data alone:

The mass hierarchy. The QED locality bound involves $\ln(\Lambda^2/m_e^2) \approx 86$. Why is $m_e/\Lambda_{\text{GUT}} \approx 10^{-14}$? This is the **hierarchy problem** — one of the deepest open questions in physics. The K-Q framework does not solve it; it **isolates** it as the single remaining freedom in the QED sector. An interesting possibility (reserved for future work) is that the Yukawa sector — which generates fermion masses — might itself be subject to a K-Q stability condition analogous to the gauge sector.

The flavor number. Why does the Standard Model have $N_f = 6$ quarks and 3 charged leptons? This is the **family problem**. The framework takes N_f as input from observation. If a K-Q stability condition constrains N_f (perhaps requiring $\beta_0 > 0$ for asymptotic freedom, which gives $N_f < 16.5$ for SU(3)), this would further narrow the ‘space of admissible Standard Models’ — another direction for future work.

The honest conclusion: **The framework derives everything that gauge group topology can determine, and clearly identifies what it cannot.** The remaining freedoms are exactly the known open problems of particle physics.

7 Discussion

We collect the key points and address potential objections.

7.1 Is the fixed-point result a tautology?

One might object that ALPHA-18 is algebraically trivial: substituting $\ln(\mu^2/\Lambda^2) = 4\pi/(\beta_0\alpha_s)$ into the bound $\alpha_s\beta_0 L/2\pi < 1$ always gives 2. This is correct at the level of algebra. The **content** is not the algebra but the interpretation:

1. In the standard framework, dimensional transmutation is a **consequence** of quantization and the locality bound is a **condition** for perturbativity. They are treated as independent statements.
2. In the K-Q framework, both are aspects of a **single structural requirement**: projection stability. The transmutation formula says ‘the coupling determines the confinement scale.’ The locality bound says ‘the confinement scale constrains the coupling.’ They are the same statement viewed from opposite ends of the self-consistency loop.

The fixed-point result is not new mathematics. It is a new **structural identification**: the self-consistency of $Q \circ K$ under radiative corrections is the physical content that both dimensional transmutation and the locality bound express.

7.2 What this paper does not claim

We do not claim:

1. That $\alpha = 1/137$ is derivable from first principles (the mass hierarchy remains open).

2. That the K-Q framework is the unique framework producing these bounds (perturbative QFT already knows the perturbativity condition).
3. That the Standard Model is the unique theory satisfying projection stability (other gauge groups and matter contents may also be admissible).

We **do** claim:

1. That the K-Q framework provides a **structural interpretation** (projection stability) that unifies the locality bound, dimensional transmutation, and the mass gap into a single self-consistency condition.
2. That this interpretation **strengthens** the mass gap reduction by promoting $\gamma > 0$ from empirical to structural.
3. That the qualitative hierarchy $\alpha_{\text{QED}} \ll \alpha_s$ is a **topological theorem** following from $\pi_3(U(1)) = 0$ versus $\pi_3(\text{SU}(3)) = \mathbb{Z}$.

8 Conclusion

The coupling constants of the Standard Model are not arbitrary numbers waiting for a deeper theory to explain. Within the K-Q projection framework, they are **projection-stability parameters** — values at which the composed map $Q \circ K$ produces observables robust under radiative corrections. The derivation chain from gauge group data to coupling bounds is:

$$G \rightarrow C_A, T_f \rightarrow \beta_0 \rightarrow \Lambda \rightarrow \text{locality bound} \rightarrow \alpha.$$

For QCD, the chain closes into a self-consistency loop whose fixed point **is** the physical coupling. The thimble-comb isomorphism further establishes that this fixed point is **unique within its kernel class** (Banach contraction) and **dynamically stable** (perturbations decay). The coupling is an attractor of the self-consistency loop, not merely a boundary value. For QED, the chain produces an upper bound, with the measured value inside by a factor reflecting the mass hierarchy.

The question ‘Why 1/137?’ decomposes into:

1. ‘Why is $C_{A(U(1))} = 0$?’ — Because the gauge group is abelian. (Answered.)
2. ‘Why is $|\beta_0| \cdot \ln(\Lambda^2/m^2) \approx 115$?’ — Because the mass hierarchy is $\sim 10^{-14}$. (Open: hierarchy problem.)

The question ‘Why 0.118?’ decomposes into:

1. ‘Why is $\beta_0 = 7$?’ — Because $C_A = 3$ and $N_f = 6$. (Answered.)
2. ‘Why does the coupling saturate the bound?’ — Because dimensional transmutation and the locality bound form a tautological loop. (Answered: fixed-point result.)

The era of treating coupling constants as unexplained parameters is approaching its end. What remains are the **structural** open problems: the mass hierarchy and the family number. These are the frontiers where the K-Q framework must next be tested.

Perhaps the deeper contribution of this paper is not the specific result about α but the recurring principle it exemplifies: **locality is not fundamental ontology but stable projected kernel behavior**. The exponential decay of Green’s functions, the spectral gap of the propagator, the finiteness of position-space correlations — these are not axioms of the theory but consequences of

projection stability. The coupling constant is the parameter that controls whether this stability holds. In this light, the question ‘Why is α small?’ becomes ‘Why does observable geometry survive quantization?’ — and the answer is: because the gauge group’s topology constrains the projection to a regime where the spectral gap is preserved. Locality is reconstructed, not assumed; and couplings are the price of that reconstruction.

8 References

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